

EXPERIMENTS

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Experiment 1: Outlier Detection in Real-World Data

Objective: Identify and analyse extreme values in datasets.

Task:

- Collect a dataset such as house prices, salaries, or transaction amounts.
- Visualize the data using box plots and scatter plots.
- Apply IQR (Interquartile Range) method to detect outliers.
- Compare results with Z-score based outlier detection.
- Analyse how outliers affect mean and standard deviation.

Objective

To identify and analyse outliers in a real-world dataset using Box Plot, Scatter Plot, IQR method, and Z-score method, and study their effect on the mean and standard deviation.

Theory

Outlier detection is the process of identifying observations that differ significantly from the rest of the dataset. These unusual values may occur due to measurement errors, data entry mistakes, or genuine rare events.

Two common methods are:

IQR Method

The Interquartile Range (IQR) measures the spread of the middle 50% of the data. • $IQR = Q3 - Q1$

• Lower Limit = $Q1 - 1.5 \times IQR$

• Upper Limit = $Q3 + 1.5 \times IQR$

Values outside these limits are considered outliers.

Z-Score Method

The Z-score measures how many standard deviations a value lies from the

$$\text{mean. } Z = \frac{x - \mu}{\sigma}$$

If $|Z| > 3$, the observation is generally treated as an outlier.

Outliers influence the mean and standard deviation more than the median and IQR. Detecting them improves the reliability of data analysis.

Dataset

Use this dataset directly inside the program.

House Price

250

270

290

300

310

330

340

360

370

390

410

430

450

470

500

520

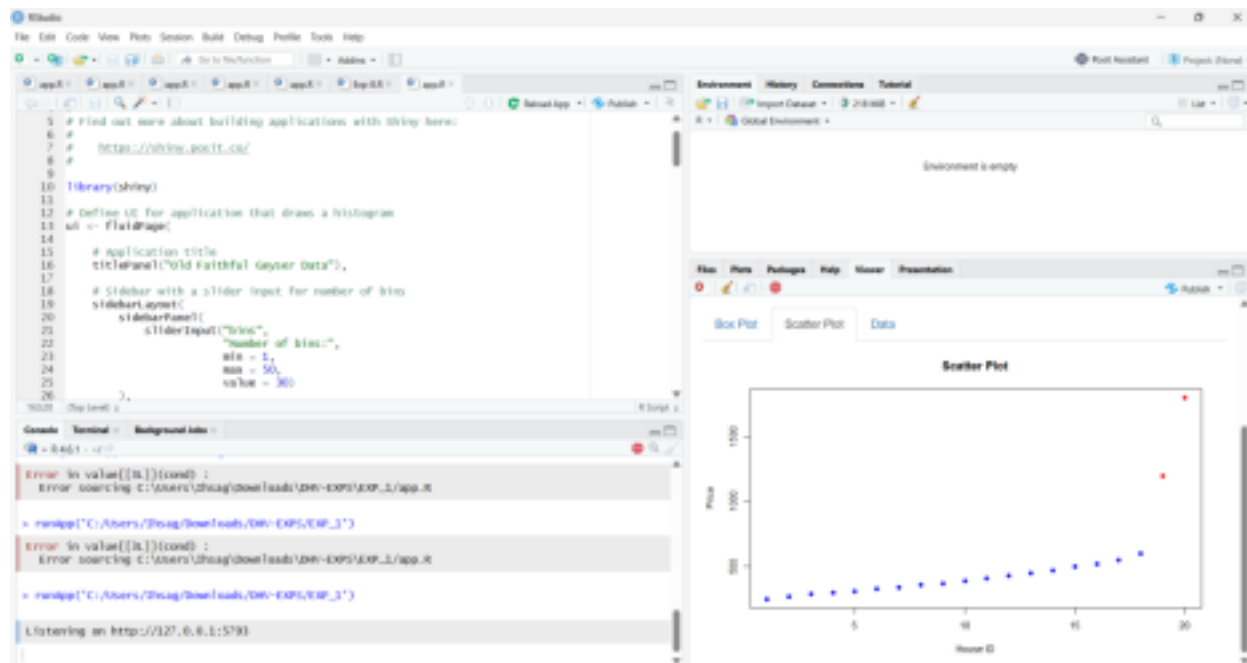
550

600

1200

1800

Output:



Interpretation

The Box Plot clearly identifies extreme house prices as outliers. The IQR and Z-score methods successfully detect abnormal observations. After removing these outliers, both the mean and standard deviation decrease significantly, indicating that extreme values strongly influence statistical measures. The analysis demonstrates the importance of outlier detection before performing further data analysis.

Result

Thus, outliers were successfully detected using the IQR and Z-score methods. Their impact on statistical measures was analyzed using visualization techniques such as Box Plot and Scatter Plot, fulfilling the objective of the experiment.

Experiment 2: Correlation and Relationship Analysis

Objective: Understand relationships between variables in multivariate data. Task:

- Use a dataset containing at least 4 numeric variables (e.g., car features like mileage, price, engine power, weight).
- Compute the correlation matrix.
- Visualize using a heatmap.
- Plot scatter plots for highly correlated pairs.
- Interpret positive, negative, and weak correlations.

Aim

To analyse the relationship between multiple numerical variables using a correlation matrix, heatmap, and scatter plot, and interpret positive, negative, and

weak correlations.

Objective

- To calculate the correlation matrix.
- To visualize relationships using a heatmap.
- To identify highly correlated variables using scatter plots.
- To understand positive, negative, and weak correlations.

Correlation analysis is a statistical technique used to measure the strength and direction of the relationship between two or more variables. The correlation coefficient (r) ranges from **-1 to +1**.

- **Positive Correlation (+1)**: Both variables increase together.
- **Negative Correlation (-1)**: One variable increases while the other decreases.
- **Zero Correlation (0)**: No relationship exists between variables.

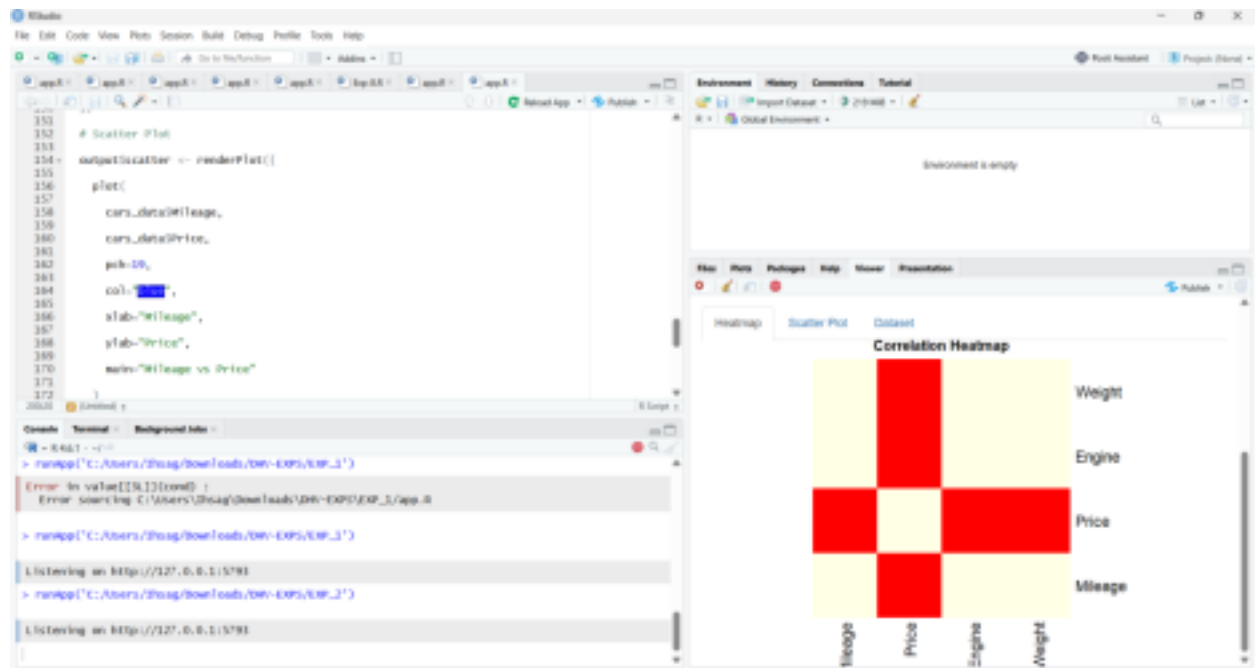
A **Correlation Matrix** displays correlation coefficients between all variables. A **Heatmap** uses different colors to represent correlation strength, making relationships easier to understand visually. A **Scatter Plot** helps identify trends, clusters, and outliers between two variables.

Dataset

Mileage Price Engine Weight

18	12	1000	850
20	11	1200	900
22	10	1200	920
24	9	1400	980
25	8.5	1500	1000
28	8	1600	1050
30	7	1800	1100
32	6.5	1800	1150
35	6	2000	1200
38	5.5	2200	1300

Output:



Experiment 3: Time Series Trend and Seasonality Analysis

Objective: Analyze temporal patterns in sequential data.

Task:

- Collect time series data (e.g., monthly sales, temperature, website traffic).
- Plot line charts to observe trends over time.
- Decompose the series into trend, seasonal, and residual components.
- Identify patterns such as seasonality or irregular fluctuations.
- Compare raw data vs smoothed moving average.

Aim

To analyze time series data by identifying trends, seasonality, and irregular variations using line charts, decomposition, and moving averages.

Objective

- To visualize monthly sales data using a line chart.
- To identify trend and seasonal components.
- To compare raw data with moving averages.
- To analyze temporal patterns in the dataset.

A **Time Series** is a sequence of observations recorded over regular intervals such as daily, monthly, or yearly.

Time series analysis helps identify:

- **Trend:** Long-term increase or decrease in data.
- **Seasonality:** Regular repeating patterns over time.
- **Residual (Irregular):** Random fluctuations not explained by trend or seasonality. •

Moving Average: Smooths short-term fluctuations and highlights long-term trends.

Time series visualization is widely used in sales forecasting, stock market analysis, weather prediction, and website traffic analysis.

Dataset **Month Sales**

Jan 120

Feb 130

Mar 125

Apr 140

May 150

Jun 165

Jul 170

Aug 180

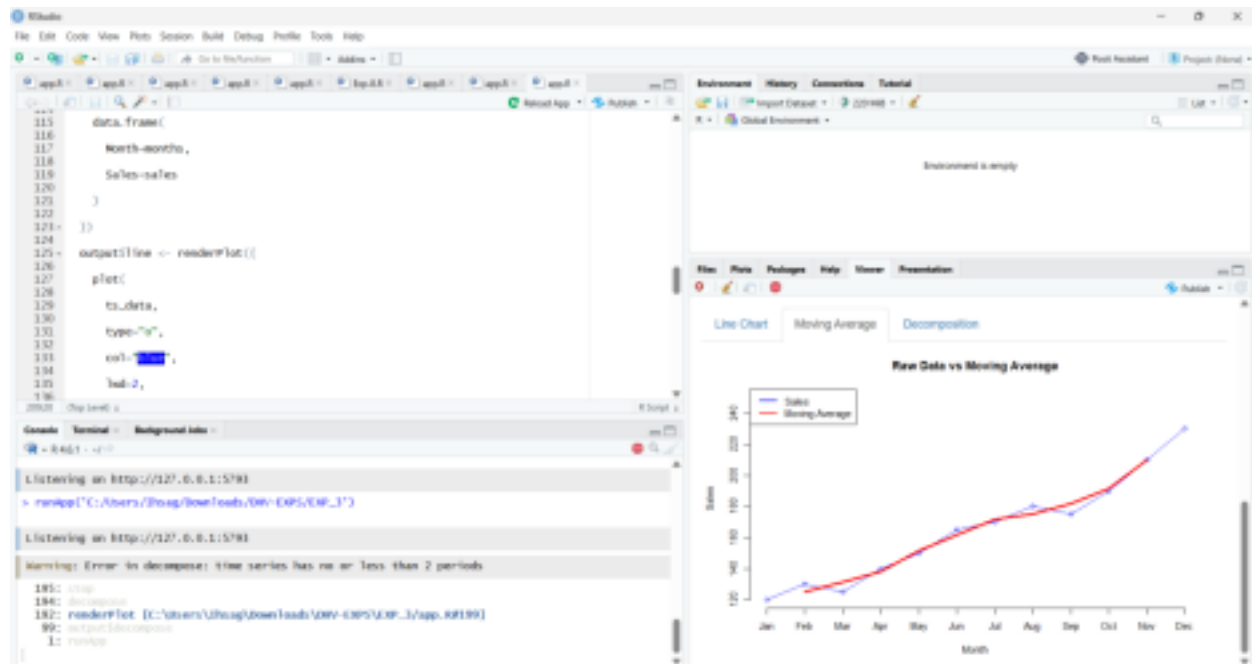
Sep 175

Oct 190

Nov 210

Dec 230

Output:



Interpretation

The line chart shows that monthly sales increase steadily throughout the year, indicating a positive trend. The moving average smooths short-term fluctuations and clearly highlights the long-term growth pattern. The decomposition graph separates the data into trend, seasonal, and random components, making it easier to understand seasonal variations and irregular fluctuations.

Result

Thus, the time series data was successfully analysed using line charts, moving averages, and decomposition techniques. The experiment effectively identified trend and seasonal patterns in the monthly sales data, achieving the objective of the experiment.